

Mairi O'Brien

POSTDOCTORAL RESEARCH FELLOW · ASTRONOMY & ASTROPHYSICS

University of Warwick, Gibbet Hill Road, Coventry, UK. CV4 7ES

☎ +44 7701 314635 | ✉ mairi.obrien1@gmail.com

Professional Experience

University of Warwick

POSTDOCTORAL RESEARCH FELLOW

- Funding: European Research Council (ERC) and Science and Technology Facilities Council (STFC)

Coventry, UK

July 2025 - present

Education

University of Warwick

PHD ASTROPHYSICS

- Thesis Title: Spectroscopic and Photometric Analysis of White Dwarfs in the Solar Neighbourhood
- Advisor: Prof. Pier-Emmanuel Tremblay
- Funding: European Research Council (ERC)

Coventry, UK

Oct 2021 - June 2025

Durham University

MPHYS PHYSICS & ASTRONOMY

- First Class Integrated Masters
- Masters Project: Optical variability properties of accreting white dwarfs observed with NASA's TESS satellite
- Advisor: Prof. Simone Scaringi

Durham, UK

Oct 2017 - July 2021

Publications

FIRST AUTHOR

5. **M. W. O'Brien**, P.-E. Tremblay, B. T. Gänsicke, M. A. Hollands, D. Koester, S. Sahu, A. Bédard, A. M. Buchan, T. Cunningham, J. H. Debes, J. J. Hermes, & P. M. Kowalski. 2026. **White dwarfs within 13 pc: Insights from ultraviolet spectroscopy**. Submitted to MNRAS. *arXiv:2607.06357*
4. **M. W. O'Brien**, D. J. Wilson, P.-E. Tremblay, B. T. Gänsicke, C. M. Byrne, F. Lagos-Vilches, J. S. Pineda, & J. Barraza-Jorquera. 2026. **Direct detections of white dwarfs in four WD+DM post-common envelope binaries within 20 pc**. MNRAS 550, 2. *arXiv:2607.11320*
3. **M. W. O'Brien**, P.-E. Tremblay, B. L. Klein, C. Melis, D. Koester, A. M. Buchan, D. Veras, & A. E. Doyle. 2025. **Characterizing planetary material accreted by cool helium atmosphere white dwarfs using an exponentially decaying disc model**. MNRAS 539, 1, 171. *arXiv:2503.05345*
2. **M. W. O'Brien**, P.-E. Tremblay, B. L. Klein, D. Koester, C. Melis, A. Bédard, E. Cukanovaite, T. Cunningham, A. E. Doyle, B. T. Gänsicke, N. P. Gentile Fusillo, M. A. Hollands, J. McCleery, I. Pelisoli, S. Toonen, A. J. Weinberger & B. Zuckerman. 2024. **The 40 pc sample of white dwarfs with Gaia**. MNRAS 527, 3, 8687. *arXiv:2312.02735*
1. **M. W. O'Brien**, P.-E. Tremblay, N. P. Gentile Fusillo, M. A. Hollands, B. T. Gänsicke, D. Koester, I. Pelisoli, E. Cukanovaite, T. Cunningham, A. E. Doyle, A. Elms, J. Farihi, J. J. Hermes, J. Holberg, S. Jordan, B. L. Klein, S. J. Kleinman, C. J. Manser, D. De Martino, T. R. Marsh, J. McCleery, C. Melis, A. Nitta, S. G. Parsons, R. Raddi, A. Rebassa-Mansergas, M. R. Schreiber, R. Silvotti, D. Steeghs, O. Toloza, S. Toonen, S. Torres, A. J. Weinberger & B. Zuckerman. 2023. **Gaia white dwarfs within 40 pc III: spectroscopic observations of new candidates in the southern hemisphere**. MNRAS 518, 2, 8687. *arXiv:2210.01608*

CO-AUTHOR

12. H. Dawson, M. Dorsch, S. Geier, J. Munday, M. Pritzkeleit, U. Heber, F. Mattig, D. Benitez-Palacios, M. Vučković, I. Pelisoli, K. Deshmukh, A. Bhat, L. Kufleitner, M. Uzundag, V. Schaffenroth, N. Reindl, R. Culpan, R. Raddi, L. Antunes Amaral, A. G. Istrate, S. Justham, R. H. Østensen, J. H. Telting, T. Steinmetz, N. Rodríguez-Segovia, P. Fernandez-Schlosser, A. Durán-Reyes, E. Arancibia-Rojas, M. Latour, G. T. Jones, **M. W. O'Brien**, S. Sahu, & A. Elms. 2026. **A 500 pc volume-limited**

sample of hot subluminous stars: II. Atmospheric parameters, mass distribution, and kinematics. A&A 707, A6, 20. *arXiv:2601.02250*

11. S. Sahu, P.-E. Tremblay, D. Koester, **M. W. O'Brien**, S. Blouin, B. T. Gänsicke & V. Fairchild. 2025. **Re-evaluating Lyman α wing opacities and the low mass-problem in cool white dwarfs.** MNRAS 544, 2, 2038. *arXiv:2511.00151*
10. T. Cunningham, P.-E. Tremblay, **M. W. O'Brien**, E. B. Bauer, M. A. Hollands, D. Koester, S. J. Kenyon, D. Charbonneau, D. Veras & M. F. Yusaf. 2025. **The dearth of high-mass hydrogen-atmosphere metal-polluted white dwarfs within 40 pc.** MNRAS 539, 3, 2021. *arXiv:2503.09734*
9. E. K. Roberts, P.-E. Tremblay, **M. W. O'Brien**, A. Bédard, T. Cunningham, C. M. Byrne & E. Cukanovaite. 2025. **Comparison of methods used to derive the Galactic star formation history from white dwarf samples.** MNRAS 538, 4, 2548. *arXiv:2502.09579*
8. P.-E. Tremblay, A. Bédard, **M. W. O'Brien**, J. Munday, A. K. Elms, N. P. Gentile Fusillo & S. Sahu. 2024. **The Gaia white dwarf revolution.** New Astronomy Reviews, vol 99. *arXiv:2402.14960*
7. A. K. Elms, N. P. Gentile Fusillo, P.-E. Tremblay, R. C. Bohlin, M. A. Hollands, S. Sahu, **M. W. O'Brien**, S. Deustua, & T. Cunningham. 2024. **A network of cooler white dwarfs as infrared standards for flux calibration.** MNRAS 534, 3, 2758. *arXiv:2410.06754*
6. J. T. Williams, B. T. Gänsicke, A. Swan, P. Izquierdo, **M. W. O'Brien**, A.-M. Cutolo, & T. Cunningham. 2024. **PEWDD: A database of white dwarfs enriched by exo-planetary material.** A&A 691, A352, 19. *arXiv:2409.16046*
5. J. Munday, I. Pelisoli, P.-E. Tremblay, T. R. Marsh, G. Nelemans, A. Bédard, S. Toonen, E. Breedt, T. Cunningham, **M. W. O'Brien**, H. Dawson. 2024. **The DBL Survey I: discovery of 34 double-lined double white dwarf binaries.** MNRAS 532, 2, 2534. *arXiv:2407.02594*
4. C. López-Sanjuan, P.-E. Tremblay, **M. W. O'Brien**, D. Spinoso, A. Ederoclite, H. Vázquez Ramió, A. J. Cenarro, A. Marín-Franch, T. Civera, J. M. Carrasco, B. T. Gänsicke, N. P. Gentile Fusillo, A. Hernán-Caballero, M. A. Hollands, A. del Pino, H. Domínguez Sánchez, J. A. Fernández-Ontiveros, F. M. Jiménez-Esteban, A. Rebassa-Mansergas, L. Schmidtobreick, R. E. Angulo, D. Cristóbal-Hornillos, R. A. Dupke, C. Hernández-Monteagudo, M. Moles, L. Sodré Jr, & J. Varela. 2024. **J-PLUS: The fraction of calcium white dwarfs along the cooling sequence.** A&A 691, A211, 16. *arXiv:2406.16055*
3. T. Cunningham, P.-E. Tremblay & **M. W. O'Brien**. 2024. **Initial-final mass relation from white dwarfs within 40 pc.** MNRAS 527, 2, 3602. *arXiv:2310.15410*
2. E. Cukanovaite, P.-E. Tremblay, S. Toonen, K. D. Temmink, C. J. Manser, **M. W. O'Brien** & J. McCleery. 2023. **Local stellar formation history from the 40 pc white dwarf sample.** MNRAS 522, 2, 1643. *arXiv:2209.13919*
1. S. Scaringi, P. J. Groot, C. Knigge, A. J. Bird, E. Breedt, D. A. H. Buckley, Y. Cavecchi, N. D. Degenaar, D. de Martino, C. Done, M. Fratta, K. Ikiewicz, E. Koerding, J.-P. Lasota, C. Littlefield, C. F. Manara, **M. O'Brien**, P. Szkody & F. X. Timmes. 2022. **Localized thermonuclear bursts from accreting magnetic white dwarfs.** Nature 604, 7906, 447. *arXiv:2204.09070*.

Awards

- 2024 **Warwick Award for Public and Community Engagement Team Award: Coding with Sophie**, University of Warwick
- 2023 **First Prize: Second-Year Physics PhD Poster Presentation**, University of Warwick
- 2023 **Excellence in Gender Equality Team Award - Highly Commended: Equitea**, University of Warwick
- 2020 **Physics Department Award for Excellence**, Durham University

Press

- 2026 **Dead stars in our cosmic backyard: scientists spot four white dwarfs hiding under our noses**, University of Warwick and University of Colorado Boulder. Reported by 23 news outlets. Lead author
- 2022 **New type of stellar explosion called micronovae discovered**, Durham University. Reported by 123 news outlets. Co-author

Telescope Time Awarded _____

- 2026 **NIRSpec and MIRI on the James Webb Space Telescope**, 80.9 hours (PI)
- 2024 **STIS on the Hubble Space Telescope**, 5 orbits (PI)
- 2023 **X-shooter on the Very Large Telescope**, 4.7 hours (PI)
- 2021 **XRT on the Neil Gehrels Swift Observatory**, 3000 s (PI)

Presentations _____

CONFERENCE TALKS & POSTERS

- 2026 **Talk**, EuroWD26 *Vienna, Austria*
- 2025 **Talk**, National Astronomy Meeting 2025 *Durham, UK*
- 2024 **Talk**, EuroWD24 *Barcelona, Spain*
- 2024 **Talk (virtual)**, Cool Stars 22 *San Diego, USA*
- 2024 **Poster**, Current Challenges in the Physics of White Dwarf Stars *Santa Fe, USA*
- 2023 **Poster**, Exoplanet Systems and Stellar Life Cycles: Late-Stage and Post-MS Systems *Aspen, USA*
- 2022 **Talk**, EuroWD22 *Tübingen, Germany*
- 2022 **Talk**, National Astronomy Meeting 2022 *Warwick, UK*

SEMINAR TALKS

- 2025 **Seminar Series**, Sheffield University *Sheffield, UK*
- 2024 **Student Discussion Talk**, Armagh Observatory *Armagh, UK*
- 2024 **Tuesday Lunch Talk**, UCLA *Los Angeles, USA*

Personal Development _____

JOURNAL REFEREEING

- 2025 – present **ApJ and ApJS**, 2 articles

TEACHING

- 2021 – 2024 **Graduate Teaching Assistant**, Physics Department, University of Warwick
- 2022 – 2024 **Undergraduate Exam Invigilator**, Physics Department, University of Warwick

OUTREACH & LEADERSHIP

- 2026 – 2027 **UK Gaia DR4 White Dwarf Workshop**, Organising Lead
- 2025 – 2026 **Warwick Mobile Planetarium**, Volunteer
- 2023 – 2025 **Conference for Undergraduate Women in Physics (CUWiP)**, Local Organising Committee Member
- 2023 – 2025 **Coding with Sophie**, Teacher
- 2023 – 2024 **Postgraduate Student-Staff Liaison Committee**, Chair
- 2022 – 2024 **Equitea Seminars**, Committee Member
- 2022 **Warwick Astronomy Knowledge Exchange (WAKE)**, Workshop Organiser

DEVELOPMENT

- 2026 **DataCamp: SQL Associate Data Analyst**, Certification
- 2024 **JWST Proposal Planning Workshop, University of Edinburgh**, Participant
- 2023 **Isaac Newton Telescope, La Palma, Spain**, Visiting Observer: 5 Nights
- 2022 **Summer Schools in Astrostatistics & Astroinformatics, Penn State University**, Virtual Participant